P8 Report

Time Complexity

Let n be the number of posts currently stored in the Max Heap, k be the number of trending post requested, and h be the number of BST's height.

For each post display, print_trending calls extract_max on the Max Heap. Because the heap is a complete binary tree, its height is $\log_2 n$, so extract_max takes O(log n).

The function also searches the BST for the username and may insert a new entry or updating an existing one. Therefore, these BST operations take O(h).

Above all, processing one post takes O(log n + h), and processing up to k posts takes:

O (k (log n + h))

For the best case, h = $\log_2 n$, the complexity is O (k log n).

For the worst case, h = n – 1, the complexity is O(kn).

Evidence of Testing

```
/Users/yihaoyu/Desktop/FIT2085/Ontrack_Files/P8/mai
Test 1 for system_efficient PASSED.
Test 2 for system_efficient PASSED.
Test 3 for system_efficient PASSED.

Process finished with exit code 0
```

```
yihaoyu@dyn-118-138-78-77 P8 % ./main test
1. Add a post
2. Get the next trending list
3. Check user's trending post
4. Test if a tree is efficient
5. See the menu again
6. Quit
Enter command (1-6):
1
Enter the username of the poster:
A
Enter the number of likes:
1
Enter command (1-6):
1
Enter the username of the poster:
B
Enter the number of likes:
2
Enter command (1-6):
3
Enter the username:
A
User does not have any trending posts
Enter command (1-6):
5
1. Add a post
2. Get the next trending list
3. Check user's trending post
4. Test if a tree is efficient
5. See the menu again
6. Quit
Enter command (1-6):
1
Enter the username of the poster:
C
Enter the number of likes:
3
Enter command (1-6):
1
Enter the username of the poster:
D
Enter the number of likes:
4
```

```
Enter command (1-6):
1
Enter the username of the poster:
E
Enter the number of likes:
5
Enter command (1-6):
1
Enter the username of the poster:
F
Enter the number of likes:
6
Enter command (1-6):
1
Enter the username of the poster:
G
Enter the number of likes:
7
Enter command (1-6):
5
1. Add a post
2. Get the next trending list
3. Check user's trending post
4. Test if a tree is efficient
5. See the menu again
6. Quit
Enter command (1-6):
2
Enter how many posts to read from the trending list:
2
Trending Post byG with 7 likes.
Trending Post byF with 6 likes.
Enter command (1-6):
4
Enter number of nodes
1
Enter node keys:
1
Tree is efficient
Enter command (1-6):
3
Enter the username:
a
User does not have any trending posts
Enter command (1-6):
3
Enter the username:
G
```

```
User's best trending post has 7 likes.
Enter command (1-6):
3
Enter the username:
F^[
User does not have any trending posts
Enter command (1-6):
3
Enter the username:
E
User does not have any trending posts
Enter command (1-6):
3
Enter the username:
F
User's best trending post has 6 likes.
Enter command (1-6):
3
Enter the username:
G
User's best trending post has 7 likes.
```

```
Enter command (1-6):
1
Enter the username of the poster:
G
Enter the number of likes:
100
Enter command (1-6):
2
Enter how many posts to read from the trending list:
1
Trending Post byG with 100 likes.
Enter command (1-6):
3
Enter the username:
G
User's best trending post has 100 likes.
```